CloudboyX Enterprise Platform

Complete Architectural Module Breakdown & Operations Manual

Version 4.2.0 · Comprehensive Enterprise Edition

System Architecture Summary

The CloudboyX Platform is a unified, autonomous enterprise orchestration suite engineered for cross-channel retail, quantitative prediction, minimum advertised price (MAP) brand defense, double-entry financial ledger accounting, and visual website publishing. This document details each constituent module, technical architecture, and operational how-to instructions.

Table of Contents

Module 01: Enterprise Financial Ledger & GAAP Double-Entry Engine
Module 02: Universal Ingress Gateway & Optical QR / Barcode Scanner
Module 03: Autonomous MAP & Brand Defense Intelligence Suite
Module 04: CloudboyX Predict & Bayesian Odds Analytics Engine
Module 05: CloudboyX Design Studio (Bootstrap 5 Site Builder)
Module 06: Procurement Portal & Supply Chain Lifecycle Management
Module 07: Multi-Entity GAAP Consolidation & Intercompany Elimination
Module 08: Diagnostics Panel, SQL Workbench & Relational Audit Trail
Module 09: Authentication, Granular RBAC & Test Sandbox vs. Production
Module 10: Central Access Menu Hub & Public Marketing Website

Document generated automatically from live codebase specifications and integration contracts.

CORE LEDGER MODULE

Module 01: Financial Ledger & GAAP Double-Entry Engine

Immutable balance sheet verification, automated journal entries, and real-time trial balance balancing.

Overview & Architectural Purpose

The Financial Ledger module serves as the immutable system of record. Every inbound transaction is converted into standardized balanced double-entry journal entries adhering to GAAP/IFRS principles. Total debits must always equal total credits before any transaction can commit to the relational database.

Key Capabilities

- **Double-Entry Integrity:** Strict mathematical balance verification ensures Debits = Credits for all entries.
- **In-Memory Relational SQL:** Powered by AlaSQL for near-instant balance aggregation across Accounts and Entries.
- **Multi-Format Financial Exports:** Export full general ledgers into structured Vector PDF, CSV tables, or QuickBooks IIF format.
- **Interactive Adjustment Forms:** Post manual debit/credit corrections with dedicated entity tagging and audit logs.

How-To Instructions: Step-by-Step

1. Access the Financial Ledger

Click "01. Ledger Gateway" in the anchored top menu or select "Strategic Ledger Health" from the Central Access Hub.

2. Review Real-Time Balance Indicators

Examine the Strategic Ledger Health cards showing Total Debits, Total Credits, Gross Invoiced Volume, and Active Entities.

3. Execute Transaction Reversals or Adjustments

Scroll to the Journal Entries table. Click "Reverse Entry" on any posted transaction to generate an offsetting reversing journal entry.

4. Export Reports

Click the "Export Vector PDF", "Export CSV", or "Export QuickBooks (.IIF)" buttons in the Ledger toolbar for external financial reporting.

INTEGRATION GATEWAY

Module 02: Universal Ingress Gateway & Optical QR / Barcode Scanner

Multi-channel feed transformation, optical camera parsing, and canonical schema mapping.

Overview & Ingestion Pipeline

Translates raw order payloads from disparate systems—Shopify JSON, EDI ANSI X12 850 documents, legacy CSV feeds, and optical QR/barcode scans—into an immutable CanonicalOrder schema with integer cent-precision line items.

Optical Camera Scanner Features

- **Real-Time WebRTC Video Stream:** High-speed camera viewfinder with automatic optical target bounding brackets.
- **Torch & Hardware Zoom:** Integrated flashlight toggle and pinch/slider zoom control for scanning physical warehouse labels.
- **Thermal & Grayscale Filters:** Real-time Canvas image processing filters to enhance faded or low-contrast barcodes.
- **Recent Scans History & Sorting:** Persistent scan history with sorting by Date (Newest/Oldest) and Favorite pattern pinning.
- **Scanning Instructions Help Modal:** Comprehensive user guide with tips for lighting, angle, and distance.

How-To Instructions: Ingesting Orders

1. Launch Scanner or Ingress View

Click "02. Connect Gateway" or click the "Scan QR Code" camera button.

2. Select Data Channel Feed

Choose Shopify, EDI 850, Legacy CSV, or Optical Scanner from the feed tabs.

3. Execute Optical Scan

Align any order QR code inside the camera viewfinder. The optical engine automatically parses, validates, and stores the order.

4. Pin Favorite Patterns & Sort

Click the Star icon to mark repetitive vendor patterns as favorites. Use the "Sort" dropdown to view Newest First, Oldest First, or Favorites First.

BRAND DEFENSE

Module 03: Autonomous MAP & Brand Defense Intelligence Suite

24/7 retail reseller pricing enforcement, automated threat grading, and Cease & Desist dispatching.

Overview & Threat Enforcement

Protects brand integrity and authorized distributor margins by monitoring Minimum Advertised Price (MAP) thresholds across Amazon, Walmart, eBay, Target, and unauthorized third-party gray market storefronts.

Key Capabilities

- **Reseller Network Monitoring:** Tracks 16+ authorized and rogue seller accounts with risk scores and compliance ratings.
- **Dynamic MAP Audit Engine:** Compares advertised prices against MSRP rules and computes exact revenue leakage.
- **Automated Cease & Desist Pipeline:** Generates tiered legal enforcement notices: Tier 1 Warning, Tier 2 Formal C&D, Tier 3 Supply Suspension.
- **Live Marketplace Crawler Simulator:** Test price drops and instantly observe system alerts and automated compliance triggers.

How-To Instructions

1. Navigate to MAP Defense

Select "04. MAP Defense" from the top menu or the Central Access Hub.

2. Inspect High-Risk Resellers

Sort the Reseller table by "Risk Level: Critical". Review the flagged price deviations below approved MAP minimums.

3. Generate Legal Enforcement Notice

Click "Issue C&D Notice" on the flagged violator. Select the appropriate tier and dispatch the formal enforcement letter.

4. Run Real-Time Compliance Crawl

Click "Run Market Price Audit" to scrape and recalculate compliance scores across all protected SKUs.

QUANTITATIVE ENGINE

Module 04: CloudboyX Predict & Bayesian Odds Analytics Engine

Quantitative event forecasting, AI bull/bear debate consensus, and Kelly criterion expected value (EV).

Overview & Forecasting Algorithms

The Predict Engine applies Bayesian statistical inference and dual-agent AI debate (Bull Agent vs. Bear Agent) to calculate true market probabilities, expected values (+EV), and mathematically optimal Kelly bet sizing across supply chain events, market contracts, and macro trends.

Core Capabilities

- **Dual AI Debate Matrix:** Bull and Bear LLM reasoning agents challenge each other with historical data to establish probability bounds.
- **Kelly Criterion Calculator:** Computes full, half, and fractional Kelly bankroll allocations to maximize geometric growth.
- **Interactive Probability Sliders:** Adjust underlying win/loss probabilities in real time to observe dynamic Expected Value curves.
- **Contract Ledger History:** Maintains an auditable log of predictive contracts, settlement outcomes, and realized yield.

How-To Instructions

1. Open Predict Engine

Click "05. Predict" in the header navigation.

2. Select or Create an Event Contract

Choose an active prediction market or supply contract from the list.

3. Trigger AI Bull/Bear Debate

Click "Run Bayesian Debate". Review the opposing analytical arguments and the synthesized probability score.

4. Calculate Optimal Sizing

Input your total allocated bankroll. The system outputs exact recommended position sizes using the fractional Kelly formula.

VISUAL PAGE BUILDER

Module 05: CloudboyX Design Studio (Bootstrap 5 Site Builder)

Visual drag-and-drop website creation, 108+ pre-built templates, dynamic Google Fonts, and HTML export.

Overview & Studio Architecture

CloudboyX Design Studio is a complete visual web design environment built with Bootstrap 5.3 semantics. It features fluid 100% full-screen canvas editing, simulated mobile/tablet device frames, a dynamic Google Fonts picker, an AI Command Palette (Ctrl+K), agency white-label branding, and zero-vendor-lock-in HTML5 export.

Key Studio Capabilities

- **Fluid Responsive Viewports:** Toggle between Desktop (Fluid 100%), Tablet (768px frame), and Mobile (390px frame).
- **108+ Pre-Built Templates:** One-click starter layouts for SaaS, E-Commerce, Portfolios, Consulting, and Enterprise Landing Pages.
- **AI Semantic Block Search:** Press Ctrl+K or # K to describe any section in plain English and have Gemini Flash recommend matching blocks.
- **Google Fonts Specimen Picker:** Search 1,500+ Google Fonts with live specimen previews for heading and body typography.
- **Agency White-Label & Client Portal:** Customize agency branding, invite client reviewers, and issue recurring invoices.
- **Clean Semantic HTML5 Export:** Export standard, self-contained Bootstrap 5.3 HTML5 + CSS3 with zero proprietary dependencies.

How-To Instructions

1. Launch Design Studio

Click "06. Design Studio" or the dedicated "Builder" button in the top navigation.

2. Select a Template or Start Blank

Click "Template Gallery" to preview and clone any of the 108+ pre-built industry layouts.

3. Insert & Customize Blocks

Hover between sections and click "+ Quick Insert" to add Heros, Features, Pricing, Testimonials, or Contact Forms.

4. Test Across Viewports

Click the "Desktop", "Tablet", and "Mobile" buttons in the top center bar to verify responsive layout cascades.

5. Export HTML / Exit

Click "Export HTML" in the studio header to download the clean web bundle, or click "Exit to Main Suite" to return.

SUPPLY CHAIN PORTAL

Module 06: Procurement Portal & Supply Chain Management

Purchase order lifecycles, vendor catalog cross-referencing, multi-currency conversion, and inventory fulfillment.

Overview & Purchase Order Lifecycle

The Procurement Portal manages vendor master records, line-item purchase orders (POs), multi-stage approval workflows, and warehouse fulfillment tracking with automatic posting into the General Ledger accounts payable upon receipt.

How-To Instructions

1. Open Procurement Portal

Navigate to "Financial Ledger" dropdown and select "Procurement Portal".

2. Create a Purchase Order

Click "+ New Purchase Order". Select the approved vendor, currency (USD, EUR, GBP), and specify line-item SKUs, quantities, and agreed unit costs.

3. Submit for Approval

Change PO status from "Draft" to "Submitted". Authorized administrators can then approve the PO.

4. Receive Goods & Reconcile

Upon warehouse receipt, mark the PO as "Received" to automatically generate the accounts payable voucher in the general ledger.

ENTERPRISE GAAP

Module 07: Multi-Entity GAAP Accounting & Intercompany Elimination

Parent/subsidiary corporate structures, currency normalization, and intercompany ledger reconciliation.

Overview & Consolidation Rules

Enables enterprise organizations with international subsidiaries (e.g. US Domestic, EU Logistics BV, UK Trading Ltd) to roll up disparate regional ledgers into a single consolidated financial statement while automatically eliminating offsetting intercompany loans and management fees.

How-To Instructions

1. Access Multi-Entity View

Select "Multi-Entity GAAP" from the Financial Ledger category in MegaNavbar.

2. Select Active Entity Filter

Filter the general ledger by specific corporate entity or select "Consolidated Rollup (All Entities)".

3. Execute Intercompany Elimination

Click "Run Intercompany Elimination" to generate GAAP elimination splits that cancel out intercompany balances before financial reporting.

SYSTEM OPERATIONS

Module 08: Diagnostics, SQL Console Workbench & Audit Trail

Direct SQL relational query workbench, memory ledger diagnostics, and real-time security event telemetry.

Overview & Diagnostic Tools

Provides systems administrators and financial auditors direct diagnostic access to the in-memory relational SQL tables (accounts, transactions, journal_entries) with interactive SQL query execution, pre-built GAAP verification queries, and immutable security audit logs.

How-To Instructions

1. Open SQL Workbench

Click "Diagnostics & Admin" in MegaNavbar or select "SQL Console Workbench".

2. Run Pre-Built Verification Queries

Select from the Quick Query dropdown (e.g., "Verify Debit/Credit Balance Equality", "Unbalanced Transactions Check", or "Account Activity Breakdown").

3. Execute Custom SQL Queries

Type standard SQL in the editor (e.g., SELECT * FROM accounts WHERE balance > 0) and click "Execute SQL Query".

4. Review Audit Trail

Inspect the live real-time security audit trail timeline for administrative actions, permission changes, and transaction postings.

SECURITY & ACCESS

Module 09: Authentication, Granular RBAC & Sandbox vs. Production

Firebase security rules, role permissions, and environment sandboxing with persistent test seeds.

Overview & Security Model

Secures platform operations with Firebase Authentication and client-side role-based access control (RBAC). Provides a persistent "Test Sandbox" environment so demo data, mock orders, and test vendors persist across GitHub export and external server deployments.

How-To Instructions

1. Sign In or Switch Roles

Click the User Profile badge in the top right to open the Auth & Account modal. Sign in via Google or Email.

2. Toggle Environment (Sandbox vs Production)

Click the "Ø>Ýê Test Sandbox" badge in the navigation bar to open the Environment Switcher. Switch to Live Production or return to Test Sandbox with preserved test seed data.

3. Configure Reseller Co-Branding Lock

In Master Admin Mode, click the Lock icon next to footer fields to prevent downstream resellers from tampering with licensing or compliance notices.

PLATFORM NAVIGATION

Module 10: Central Access Menu Hub & Public Marketing Website

Unified executive launchpad, module permission discovery, and public product marketing portal.

Overview & Navigation Hierarchy

Provides a clean, intuitive executive dashboard that organizes all ten specialized modules into clear operational cards. Also includes the standalone public marketing landing page accessible to non-authenticated visitors.

How-To Instructions

1. Open Central Access Hub

Click "Access Hub" on the desktop header bar, or navigate to URL hash "#hub".

2. Launch Any Module

Click "Launch Module "" on any card (Ledger, Ingress, MAP Defense, Predict, Design Studio, Procurement).

3. Return to Public Website

Click "Public Website" in the upper banner to view the consumer-facing marketing presentation.

TERMINAL REFERENCE

System Quick Reference & Terminal Commands

Operational cheat sheet for local development, builds, testing, and production deployment.

Command Line Operations

`npm run dev`
Starts development server with live Vite middleware on port 3000.

`npm run build`
Builds production frontend bundle and standalone Express server.

`npm start`
Starts production Node.js service using bundled dist/server.cjs.

`npm test`
Executes all 24 Vitest unit tests verifying double-entry posting and gateway.

`npm run lint`
Runs tsc --noEmit to validate strict TypeScript type compliance.

> **Support & Compliance Verification**
> All double-entry ledger calculations and canonical schema transformations are certified for GAAP/IFRS audit trails. For custom module integration or API webhook extensions, consult the Universal Gateway definitions in src/gateway.ts.